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Water World Magnetic Fields: Plausibility and Consequences

Water worlds, first proposed in 2004, are planets of size between Earth and Neptune that consist of up to 50 mass percent water. They orbit within their star's habitable zone and have a liquid ocean that covers their entire surface, with no dry land. Their existence is still debated, as there is currently no way to differentiate between a water world and a gas dwarf. They would have different compositions, but the same measurable mass and radius. One unexplored disambiguation method is the presence of a magnetic field on a planet in question. The presence and strength of a magnetic field depends partially on the planet's composition. The measurement of exoplanet magnetic fields is more feasible than other methods such as direct visual observation, and when combined with data obtained by other methods, a clearer picture of the interior composition of an exoplanet is possible than ever before achieved.

Among the population of over 6,000 confirmed exoplanets is a radius valley, where there are few planets with radii between 1.5 and 2 times the radius of Earth (Fulton et al. 1). Planets around this size are divided into two classes: super-Earths and sub-Neptunes, depending on which side they fall on. Explaining this valley is the subject of much research. Of particular interest is how planets are formed. The bimodal distribution of radii implies that there are two distinct types of planets, which means they either form differently from the start, or undergo some process that changes one of them. Evidence for one of these populations being composed of different proportions of materials would support the hypothesis that they form differently from the start.

The prevailing theory to explain the radius valley says that both classes of planets are fundamentally formed in the same way, the difference is that planets above a tipping point of mass can retain their gaseous atmosphere, resulting in a higher radius, while planets below that point do not have sufficient gravity to retain their atmospheres, which escape and leave only the smaller rocky core (Rogers et al. 1). A parallel theory that complicates this model is water worlds. First proposed by Leger et al in 2004, “ocean planets,” also known as water worlds, or hycean planets, are planets that formed outside the frost line, where volatiles such as water exist as solid ice in the protoplanetary disk. They are incorporated into young planets, resulting in planets with a high water mass percent, typically estimated to be between 20% and 50% (Barros, S. C. C. et al. 11). Add something about water evaporating that would change the atmospheric composition and density.

Water worlds would have a smaller metal core and silicate mantle compared to gas dwarfs, made up for by thick layers of ice and liquid water such that their bulk densities are equal. This ambiguity motivates the search for additional observational constraints, such as magnetic fields, that could provide greater insight into planetary compositions. Magnetic fields are complex and depend on a variety of factors, many of which are currently unknown even for Earth. However, it is precisely because getting a clear picture of a planetary magnetic field requires “understanding of the history and material properties of planets” that we could also use observations of a magnetic field to reverse disambiguate those properties (Stevenson 652). The presence of a magnetic field of a planet could help inform researchers about its composition and evolution.

In addition to the dynamo, a magnetic field can be generated via induction effect from a salty water ocean, as is believed to be the case on Europa and Callisto (Stevenson 653). Induced magnetic fields are created not from a planet’s own internal forces, but from interactions between the flow of a conductive component such as saltwater with its host’s magnetic field. The strength and existence of induced fields thus depends on the field of the host star. Because it varies with the

variation of the star, it will be more difficult to detect than a more stable dynamo-generated field.

The magnetic field on Europa supported the claim that there was a subsurface ocean when it was first observed (Kuskov & Kronrod 551). The variation of magnetic fields on these satellites, and the effects they induce on their host planets, is analogous to that of planets and stars. They could be studied to better understand how bodies with a high water content relate to magnetic fields.

When a dynamo stops convecting, the planet's magnetic field does not stop instantly. Instead, it diffuses over time according to the rate of magnetic diffusion (Stevenson 653). For a Sub-Neptune, the order of magnitude of years to diffuse would be similar to Earth, which is 10^4 (Brain et al. 3). The inward migration of a water world, or any planet of a similar size, is also 10^4 , which is “much shorter than... estimated planetary formation timescales” (Terquem et al. 329). Because both processes happen so quickly, and concurrently with planet formation, it is possible that they influence each other. From there, it can go on to interact with the sub-Alfvénic wind of its host star, which would cause it to migrate even closer (out of the habitable zone) and become tidally locked with the star, so that the same side of the planet is always facing the star (Brain et al. 29).

The lack of magnetic field would have a significant effect on the atmosphere of a water world. Looking at a water world that has arrived at an orbit within the temperature habitable zone of its host star, the rate of irradiation of whatever atmosphere it has will depend on the type of star it orbits. Most stars in the milky way are M-dwarfs, a type of star that is cooler and smaller than the Sun and is characterized by “powerful magnetic dynamos and resulting X-ray to UV (X-UV) activity” (Engle 1). Their lower temperature means that the habitable zone, where liquid water can exist, is much closer. Because the magnetic field of water worlds likely survives until they reach this distance, it would interact with the magnetic field of the star. This would cause the planet to become tidally locked, and to move even closer due to the exchange of angular momentum (Brain et al. 29). Tidally locked planets have a high temperature contrast between their day and night side and will

have more extreme weather events due to atmospheric convection. The atmospheres of water worlds are thus in triple jeopardy: they face more overall radiation, they have no magnetic field to protect themselves, and they are likely to become tidally locked.

Water world atmospheres at the time of planet formation would be comprised of 90% water, 5% ammonia, and 5% carbon dioxide (Leger et al. 502). Ammonia “is very sensitive to UV,” and around a sun-like planet at an Earth like orbit would be photoevaporated or converted to molecular nitrogen and hydrogen in less than 2 million years (Leger et al. 502). The higher amount of UV radiation emitted by M-dwarf stars would shorten that period. According to Leger et al, hydrogen is generally susceptible to hydrodynamical escape, and Brain et al. determine that a lack of magnetic field likely increases this rate of escape. While this escape would sweep away some of the heavier elements, it would be primarily hydrogen and helium that are removed (Leger et al. 502). This would leave higher concentrations of carbon, nitrogen, and oxygen in the atmosphere. During periods of intense radiation such as stellar flares, “production of nitrogen oxides should increase,” compounded by the higher ratio of these molecules, causing a “corresponding decrease in oxide molecules, including ozone” (Brain et al. 27). Ozone protects the surface of Earth from dangerous UV radiation. The surface of water worlds would thus be inhospitable to life as we know it.

Overall, Brain et al. conclude that the lack of a magnetic field does not necessarily influence atmospheric escape in the way once predicted, and there are many confounding factors at play. However, given the initial composition of water worlds, and the high radiation they receive from their close orbit around M-dwarfs, it is likely that the lack of a long-lived magnetic field contributes to atmosphere loss.

There are other factors that would allow water worlds to be habitable. While the lack of a magnetic field typically leads to higher irradiation of protective molecules in the atmosphere, life could still exist deeper in the ocean. Water absorbs light energy and harmful radiation the same way

an atmosphere does. This would form a goldilocks depth in the ocean where life would be protected from harmful radiation, while still being close enough to the surface to receive some light energy. In Earth's oceans, UVB radiation levels decrease faster than visible light (Dunne and Brown 113). This means that organisms at the correct depth receive energy in the visible light spectrum without being exposed to UVB radiation.

Supplying nutrients to this goldilocks depth will be the primary challenge. Since the oceans of a water world would be up to 100km deep, and temperature and pressure at the bottom would be immense and impossible for organisms to traverse, they would rely on nutrients being brought up to their habitable level (Leger et al. 503). The high pressure at the ocean-silicate boundary may supply a mechanism to bring those nutrients up. Leger et al. note that due to the low thermal conductivity of high-pressure ice, the only way for heat to escape from the core to the surface would be via “melting of ice and migration of the melt towards the upper ocean” (501). The convection of solid ice between the silicate mantle and liquid ocean floor, and subsequent melting and rising of that ice, could bring up elements necessary for life.

Detecting the magnetic field of Sub-Neptune exoplanets is just within reach, and confirmation of one is a worthwhile near-term goal. Possible observations have been noted, such as YZ Ceti b and τ Boötis b, but consensus has not yet been achieved (Pineda et al. 1; Turner et al. 1). This is because these observations rely on seeing how the planet affects the star's magnetic field and measuring that. Confirming that those observations are caused by a planet and not some other force from the star itself require long term predictable periodic observations, which requires “significant investment of observing resources as well as a mature understanding of stellar flare behaviors” (Brain et al. 16).

Developing the technology to measure exoplanetary magnetic fields more easily would not only enhance our ability to find earth-like planets, it would also help differentiate between water

worlds and gas dwarfs. The degeneracy between them arises because of them both having the same density. Either a small amount of heavy core with a large amount of medium-weight water, or a larger heavy core with a moderate amount of very light gas. Both result in the same mass and radius measurements using Radial Velocity and Transit methods to measure each. Adding a third measurement, the magnetic obstacle size via measuring star-planet interactions as described by Brain et al, would thoroughly distinguish between the two (16). As discussed, it is unlikely for a water world to have a magnetic field. Therefore, its obstacle size would be equal to its radius. The magnetic field of a gas dwarf would extend the obstacle size measurably beyond its radius.

Magnetic fields of exoplanets remain a promising frontier of near-term research. The technology to reliably measure them is close to being realized, and upcoming missions like the ARIEL mission and Habitable Worlds Observatory will bring even deeper understanding of exoplanet magnetic fields and formation in general. The more that is learned about exoplanets, the more we must appreciate Earth for its seeming rarity and conduciveness to life. It remains the only planet amenable to life, and even though technology is advancing rapidly, it is likely that Earth will remain the only planet we have, so we must take care to ensure it remains hospitable to life.

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